CLASS 10 · CBSE · NCERT MATHEMATICS · FORMULA BANK

Chapter 4: Quadratic Equations

Every formula and result of the chapter on one sheet

1	Standard form	$ax^2 + bx + c = 0, a \neq 0$
2	Quadratic formula	$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$
3	Discriminant	$D = b^2 - 4ac$
4	Two distinct real roots	$D > 0$
5	Two equal real roots	$D = 0$; each root is $-\frac{b}{2a}$
6	No real roots	$D < 0$
7	Real roots exist	$D \geq 0$
8	Sum of the roots	$\alpha + \beta = -\frac{b}{a}$
9	Product of the roots	$\alpha\beta = \frac{c}{a}$
10	Equation from its roots	$x^2 - (\alpha + \beta)x + \alpha\beta = 0$
11	Splitting the middle term	Two numbers with sum b and product ac
12	Consecutive integers	$x, x + 1$; consecutive even or odd integers: $x, x + 2$
13	Time, distance and speed	$\text{time} = \frac{\text{distance}}{\text{speed}}$